Data Set Descriptions

In this project I queried 2 different datasets, intel.device_data and intel.impact_data, that I joined together for my analysis. Here you'll find the data dictionary for each dataset.

intel.device_data

- device_id: Unique identifier for each repurposed device
- device_type: Type of device, values are either "Laptop" or "Desktop"
- model_year: The year the device was manufactured (e.g., 2018, 2019, etc.)

intel.impact_data

- impact_id: Unique identifier for the repurposed device's impact record (e.g., "LP20NA141592")
- device_id: Unique identifier linking the impact record to a specific device in the intel.device_data table
- usage_purpose: The specific purpose for which the device is being repurposed, values are Education & Digital Literacy, Corporate & Enterprise, Government & Public Sector, Environmental Sustainability Programs, and Social Impact & Non-Profit
- power_consumption: Power consumption of the device in watts (W) when in use (e.g., 50W, 75W)
- energy_savings_yr: Estimated energy savings per device per year when repurposed compared to a new device, measured in kilowatt-hours (kWh)
- co2_saved_kg_yr: Estimated CO2 emissions saved per device per year from manufacturing a new device, measured in kilograms (kg).
- recycling_rate: The percentage of the device that is recyclable (e.g., 80%, 90%).
- region: The geographical region where the device was repurposed, values are "North America", "Europe", and "Asia"

Task 1: Join device_data and impact_data

Analysis:
Joining the device and impact data gives a complete dataset that includes device type, model year, usage purpose, power consumption, energy savings, CO_2 saved, recycling rate, and region. This sets the foundation for all further analysis.

Task 1: Add device_age column

Analysis:
Adding device_age shows how old each device is, helping us understand which devices are being repurposed and estimate potential energy and CO_2 savings. For example, a 2019 device is 5 years old, indicating mid-life devices are frequently repurposed.

device_id	device_type	model_year	impact_id	usage_purpose	power_consumption	energy_savings_yr	co2_saved_kg_yr	recycling_rate	region	device_age
LP22AS72961	Laptop	2022	LP22AS7296100004	Corporate & Enterprise	47	25.8	15.5	0.87	Asia	2
LP20EU11904	Laptop	2020	LP20EU1190400006	Education & Digital Literacy	51	24.2	6.1	0.87	Europe	4
LP21NA26999	Laptop	2021	LP21NA2699900006	Corporate & Enterprise	47	5.6	2.3	0.87	North America	3
DT20AS66801	Desktop	2020	DT20AS6680100007	Education & Digital Literacy	78	6.6	4.0	0.98	Asia	4
LP23NA40282	Laptop	2023	LP23NA4028200010	Corporate & Enterprise	43	26.4	10.6	0.87	North America	1
DT22NA80684	Desktop	2022	DT22NA8068400012	Corporate & Enterprise	79	24.7	9.9	0.94	North America	2
LP21EU87824	Laptop	2021	LP21EU8782400013	Corporate & Enterprise	52	37.9	9.5	0.88	Europe	3
LP19AS99290	Laptop	2019	LP19AS9929000022	Education & Digital Literacy	54	58.6	35.2	0.89	Asia	5
LP22NA56098	Laptop	2022	LP22NA5609800023	Corporate & Enterprise	45	23.9	9.6	0.87	North America	2
LP19NA31352	Laptop	2019	LP19NA3135200025	Education & Digital Literacy	55	28.9	11.6	0.89	North America	5
LP20NA34864	Laptop	2020	LP20NA3486400026	Education & Digital Literacy	53	23.0	9.2	0.88	North America	4
DT20NA55932	Desktop	2020	DT20NA5593200026	Corporate & Enterprise	82	38.3	15.3	0.96	North America	4
LP21AS53250	Laptop	2021	LP21AS5325000027	Education & Digital Literacy	50	32.6	19.6	0.89	Asia	3
LP23EU99312	Laptop	2023	LP23EU9931200033	Corporate & Enterprise	45	18.0	4.5	0.89	Europe	1
LP22NA92251	Laptop	2022	LP22NA9225100034	Corporate & Enterprise	48	47.1	18.8	0.87	North America	2
LP21NA30071	Laptop	2021	LP21NA3007100034	Corporate & Enterprise	52	34.9	14.0	0.88	North America	3
LP19AS25034	Laptop	2019	LP19AS2503400036	Corporate & Enterprise	53	23.3	14.0	0.88	Asia	5
LP22EU39620	Laptop	2022	LP22EU3962000037	Corporate & Enterprise	46	13.8	3.5	0.87	Europe	2
LP22AS53174	Laptop	2022	LP22AS5317400037	Education & Digital Literacy	46	-4.9	-2.9	0.88	Asia	2
LP22AS21742	Laptop	2022	LP22AS2174200043	Education & Digital Literacy	46	34.0	20.4	0.86	Asia	2
DT21NA35986	Desktop	2021	DT21NA3598600046	Government & Public Sector	80	31.0	12.4	0.94	North America	3
DT21AS63017	Desktop	2021	DT21AS6301700047	Government & Public Sector	81	28.4	17.0	0.94	Asia	3
DT20NA64068	Desktop	2020	DT20NA6406800046	Corporate & Enterprise	80	19.2	7.7	0.96	North America	4

Task 1: Add device_age_bucket column

Analysis

Categorizing devices into newer, mid-age, and older allows easier aggregation and analysis. It shows trends in energy and CO_2 savings by age group, helping identify which age brackets provide the most impact.

device_id	device_type	model_year	impact_id	usage_purpose	power_consumption	energy_savings_yr	co2_saved_kg_yr	recycling_rate	region	device_age	device_age_bucket
LP22NA30289	Laptop	2022	LP22NA3028902256	Education & Digital Literacy	48	4.9	1.9	0.89	North Americ	2	newer
DT18AS72788	Desktop	2018	DT18AS7278801237	Corporate & Enterprise	84	84.8	50.9	0.95	Asia	6	mid-age
LP23NA22027	Laptop	2023	LP23NA2202701237	Corporate & Enterprise	45	22.4	9.0	0.86	North Americ	1	newer
LP20EU13088	Laptop	2020	LP20EU1308801238	Education & Digital Literacy	53	26.0	6.5	0.87	Europe	4	mid-age
DT22EU83610	Desktop	2022	DT22EU8361001240	Corporate & Enterprise	77	10.5	2.6	0.94	Europe	2	newer
DT20NA18184	Desktop	2020	DT20NA1818401241	Corporate & Enterprise	80	27.2	10.9	0.95	North Americ	4	mid-age
LP19EU72320	Laptop	2019	LP19EU7232001242	Education & Digital Literacy	52	51.8	12.9	0.87	Europe	5	mid-age
LP22NA45129	Laptop	2022	LP22NA4512901243	Corporate & Enterprise	46	7.5	3.0	0.87	North Americ	2	newer
LP22AS93395	Laptop	2022	LP22AS9339501244	Corporate & Enterprise	46	22.7	13.6	0.88	Asia	2	newer
DT17AS10160	Desktop	2017	DT17AS1016001247	Corporate & Enterprise	86	62.8	37.7	0.94	Asia	7	older
DT22NA33399	Desktop	2022	DT22NA3339901249	Corporate & Enterprise	78	7.5	3.0	0.96	North Americ	2	newer
DT20AS67209	Desktop	2020	DT20AS6720901250	Education & Digital Literacy	81	42.8	25.7	0.97	Asia	4	mid-age
LP19NA76792	Laptop	2019	LP19NA7679201253	Corporate & Enterprise	53	37.6	15.0	0.87	North Americ	5	mid-age
LP21NA36311	Laptop	2021	LP21NA3631101253	Education & Digital Literacy	52	49.9	20.0	0.89	North Americ	3	newer
LP21NA62353	Laptop	2021	LP21NA6235301253	Education & Digital Literacy	49	30.8	12.3	0.88	North Americ	3	newer
DT17NA30303	Desktop	2017	DT17NA3030301256	Education & Digital Literacy	86	17.5	7.0	0.95	North Americ	7	older
LP23AS83877	Laptop	2023	LP23AS8387701259	Education & Digital Literacy	44	15.2	9.1	0.87	Asia	1	newer
LP20NA87065	Laptop	2020	LP20NA8706501261	Corporate & Enterprise	52	5.9	2.4	0.88	North Americ	4	mid-age
DT17NA53322	Desktop	2017	DT17NA5332201264	Government & Public Sector	84	18.4	7.4	0.94	North Americ	7	older
LP22NA24736	Laptop	2022	LP22NA2473601264	Government & Public Sector	47	16.7	6.7	0.88	North Americ	2	newer
DT20AS23425	Desktop	2020	DT20AS2342501264	Government & Public Sector	82	21.0	12.6	0.92	Asia	4	mid-age
LP20NA70286	Laptop	2020	LP20NA7028601265	Corporate & Enterprise	48	17.2	6.9	0.86	North Americ	4	mid-age
DT16AS74060	Desktop	2016	DT16AS7406901267	Education & Digital Literacy	90	37.7	22.6	0.97	Asia	8	older

Task 2: Aggregate overall impact

Analysis:

- Total devices repurposed: 601,740
- Average device age: 3.52 years
- Average energy savings per device: 25.74 kWh
- Total CO_2 saved: 6,768 tons

This is equivalent to powering ~1,420 U.S. households for a year or removing ~1,470 cars from the road. The program significantly reduces energy consumption and greenhouse gas emissions.

total_devices	avg_device_age	avg_energy_savings_kwh	total_co2_saved_tons
601740	3.5229700535114834	25.7420925981320836	6768.4206000000000000

Task 3: Device Type Analysis

Analysis:

- Laptops (408,064 devices) slightly outperform desktops (193,676 devices) in energy savings (~25.80 kWh vs 25.62 kWh) and CO_2 (~0.01128 vs 0.01119 tons per device).
- Laptops are less energy-efficient when manufactured new, so repurposing them avoids higher energy and CO_2 emissions per unit .

device_type	total_devices	avg_energy_savings_kwh	avg_co2_saved_tons
Desktop	193676	25.6162931906896053	0.01119045725851421960
Laptop	408064	25.8017997176913425	0.01127543130489335010

Task 3: Device Age Analysis

Analysis:

- Mid-age devices (3–6 years, 264,310 units) are most numerous and provide moderate energy (32 kWh) and CO_2 (0.014 tons) savings.
- Older devices (>6 years, 20,239 units) save the most energy (48 kWh) and CO_2 (0.021 tons) per device but are fewer.
- Newer devices (~317,191 units) are numerous but provide lower per-device savings (~19 kWh, 0.0083 tons).

device_age_bucket	total_devices	avg_energy_savings_kwh	avg_co2_saved_tons
mid-age	264310	32.0414842419885740	0.01400321629904279070
newer	317191	19.0713276858422843	0.00832964270739081500
older	20239	48.0215722120658135	0.02100611690300904190

Task 3: Regional Analysis

Analysis:

- North America has the most devices (299,478), Asia has the highest per-device CO_2 savings (0.0155 tons), and Europe has fewer devices and lower per-device CO_2 (0.0064 tons).
- Repurposing in high carbon-intensity regions maximizes environmental impact .

region	total_devices	avg_energy_savings_kwh	avg_co2_saved_tons
Asia	192881	25.7886774747123874	0.01547317413327388390
Europe	109381	25.7486089905925161	0.00643717556065495840
North America	299478	25.7097092273889902	0.01028400950988052540

Task 4: Optimizing Repurposing Strategy

Analysis:

- Laptops provide ~68% of energy and CO_2 savings in all regions, desktops ~32%.
- Focus on **laptops in Asia and North America** for maximum impact.
- Adding **cost per device** can prioritize devices with the highest savings per dollar, balancing sustainability and budget .

region	device_type	total_devices	avg_energy_savings_kwh	avg_co2_saved_tons	pct_energy_savings_by_type	pct_co2_savings_by_type
Asia	Desktop	61892	25.6795627867898921	0.015407618108964001	31.95241016151134609900	31.95222901882481220400
Asia	Laptop	130989	25.8402339127713014	0.015504149203368221	68.04758983848865390100	68.04777098117518779600
Europe	Desktop	35156	25.6104363408806463	0.006402545795881215	31.96839052401700520300	31.96795874249951747200
Europe	Laptop	74225	25.8140532165712361	0.006453577635567531	68.03160947598299479700	68.03204125750048252800
North America	Desktop	96628	25.5778987457051786	0.010231266299623297	32.10005418149453828300	32.09999653877516669200
North America	Laptop	202850	25.7724974118807000	0.010309133842740941	67.89994581850546171700	67.90000346122483330800